No.	Status	Date
LNDE-RPT-96-014-DS-8-TG-101	Issued	1996-03-08

UT Thickness Data Sheet - 8-TG-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 8-TG-101 located within the ATU-100. The objective is to establish baseline wall thickness data for this treated gas line prior to the unit entering service.

Grace Okafor, acting as the UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., performed the measurements and is responsible for the accuracy of the recorded data.

Technical Basis

All thickness measurements were conducted in accordance with the site NDE procedure QSCP-NDT-001 Rev 0. The inspection methodology adheres to the requirements specified in ASTM E797.

The data collected for this line is integrated into the broader baseline survey documented in LNDE-RPT-96-014, which covers 118 CMLs for the ATU-100.

Line Specifications

Line 8-TG-101 is constructed from A106 Gr.B with a nominal outer diameter of 219.1 mm and a nominal wall thickness of 8.18 mm. The design limits for this line are 1896 kPa(g) and 60 °C.

The minimum allowable thickness (t-min) for this line is 4.06 mm. Any reading recorded below this value shall be processed as an NCR.

Inspection Interval and Classification

Based on the service of treated gas, 8-TG-101 is designated as Class 2. The maximum allowable interval for thickness measurements is 10 years, with external visual inspections required every 5 years.

Future inspection dates and remaining life calculations will be determined by comparing subsequent readings against the baseline established by Grace Okafor, using the larger of the long-term or short-term corrosion rates.

Thickness readings, measured 1996-03-08

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	—	—	7.12	—	—	—	2001-03-08	
6-SG-101-02	7.11	3.56	—	—	7.05	—	—	—	2001-03-08	
6-SG-101-03	7.11	3.56	—	—	7.08	—	—	—	2001-03-08	
6-SG-101-04	7.11	3.56	—	—	7.09	—	—	—	2001-03-08	
6-SG-101-05	7.11	3.56	—	—	7.14	—	—	—	2001-03-08	
6-SG-101-06	7.11	3.56	—	—	7.07	—	—	—	2001-03-08	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	—	—	7.14	—	—	—	2001-03-08	
6-SG-102-02	7.11	3.56	—	—	7.11	—	—	—	2001-03-08	
6-SG-102-03	7.11	3.56	—	—	7.04	—	—	—	2001-03-08	
6-SG-102-04	7.11	3.56	—	—	6.99	—	—	—	2001-03-08	
6-SG-102-05	7.11	3.56	—	—	7.13	—	—	—	2001-03-08	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	—	—	8.20	—	—	—	2006-03-08	
8-TG-101-02	8.18	4.06	—	—	8.20	—	—	—	2006-03-08	
8-TG-101-03	8.18	4.06	—	—	8.23	—	—	—	2006-03-08	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-04	8.18	4.06	—	—	8.10	—	—	—	2006-03-08	
8-TG-101-05	8.18	4.06	—	—	8.22	—	—	—	2006-03-08	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	
4-LA-101-02	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-101-03	6.02	3.18	—	—	5.96	—	—	—	2006-03-08	
4-LA-101-04	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-101-05	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-101-06	6.02	3.18	—	—	6.06	—	—	—	2006-03-08	
4-LA-101-07	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	—	—	6.02	—	—	—	2006-03-08	
4-LA-102-02	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-102-03	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-102-04	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	
4-LA-102-05	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	
4-LA-102-06	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	
4-LA-102-07	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	—	—	7.02	—	—	—	2006-03-08	
6-LA-103-02	7.11	3.56	—	—	7.19	—	—	—	2006-03-08	
6-LA-103-03	7.11	3.56	—	—	7.08	—	—	—	2006-03-08	
6-LA-103-04	7.11	3.56	—	—	7.17	—	—	—	2006-03-08	
6-LA-103-05	7.11	3.56	—	—	7.19	—	—	—	2006-03-08	
6-LA-103-06	7.11	3.56	—	—	7.16	—	—	—	2006-03-08	
6-LA-103-07	7.11	3.56	—	—	7.18	—	—	—	2006-03-08	
6-LA-103-08	7.11	3.56	—	—	7.04	—	—	—	2006-03-08	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	—	—	8.52	—	—	—	2006-03-08	
4-RA-101-02	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-101-03	8.56	3.81	—	—	8.61	—	—	—	2006-03-08	
4-RA-101-04	8.56	3.81	—	—	8.63	—	—	—	2006-03-08	
4-RA-101-05	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-101-07	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-101-08	8.56	3.81	—	—	8.68	—	—	—	2006-03-08	
4-RA-101-09	8.56	3.81	—	—	8.57	—	—	—	2006-03-08	
4-RA-101-10	8.56	3.81	—	—	8.67	—	—	—	2006-03-08	
4-RA-101-11	8.56	3.81	—	—	8.68	—	—	—	2006-03-08	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-102-02	8.56	3.81	—	—	8.57	—	—	—	2006-03-08	
4-RA-102-03	8.56	3.81	—	—	8.56	—	—	—	2006-03-08	
4-RA-102-04	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-102-05	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	
4-RA-102-06	8.56	3.81	—	—	8.51	—	—	—	2006-03-08	
4-RA-102-07	8.56	3.81	—	—	8.58	—	—	—	2006-03-08	
4-RA-102-08	8.56	3.81	—	—	8.49	—	—	—	2006-03-08	
4-RA-102-09	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	—	—	11.00	—	—	—	2006-03-08	
6-RA-103-02	10.97	4.57	—	—	11.09	—	—	—	2006-03-08	
6-RA-103-03	10.97	4.57	—	—	10.99	—	—	—	2006-03-08	
6-RA-103-04	10.97	4.57	—	—	10.98	—	—	—	2006-03-08	
6-RA-103-05	10.97	4.57	—	—	10.95	—	—	—	2006-03-08	
6-RA-103-06	10.97	4.57	—	—	11.01	—	—	—	2006-03-08	
6-RA-103-07	10.97	4.57	—	—	10.99	—	—	—	2006-03-08	
6-RA-103-08	10.97	4.57	—	—	10.97	—	—	—	2006-03-08	
6-RA-103-09	10.97	4.57	—	—	11.01	—	—	—	2006-03-08	
6-RA-103-10	10.97	4.57	—	—	11.02	—	—	—	2006-03-08	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	
3-RV-101-02	7.62	3.56	—	—	7.57	—	—	—	2001-03-08	
3-RV-101-03	7.62	3.56	—	—	7.52	—	—	—	2001-03-08	
3-RV-101-04	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	
3-RV-101-05	7.62	3.56	—	—	7.67	—	—	—	2001-03-08	
3-RV-101-06	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	—	—	5.53	—	—	—	2001-03-08	
2-RV-102-02	5.54	2.79	—	—	5.59	—	—	—	2001-03-08	
2-RV-102-03	5.54	2.79	—	—	5.48	—	—	—	2001-03-08	
2-RV-102-04	5.54	2.79	—	—	5.58	—	—	—	2001-03-08	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	—	—	5.48	—	—	—	2006-03-08	
3-RF-101-02	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-03	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-04	5.49	2.41	—	—	5.46	—	—	—	2006-03-08	
3-RF-101-05	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-06	5.49	2.41	—	—	5.52	—	—	—	2006-03-08	
3-RF-101-07	5.49	2.41	—	—	5.42	—	—	—	2006-03-08	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	—	—	3.92	—	—	—	2006-03-08	
2-RF-102-02	3.91	2.16	—	—	3.88	—	—	—	2006-03-08	
2-RF-102-03	3.91	2.16	—	—	3.93	—	—	—	2006-03-08	
2-RF-102-04	3.91	2.16	—	—	3.87	—	—	—	2006-03-08	
2-RF-102-07	3.91	2.16	—	—	3.96	—	—	—	2006-03-08	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-104-02	6.02	3.18	—	—	6.03	—	—	—	2006-03-08	
4-LA-104-03	6.02	3.18	—	—	5.96	—	—	—	2006-03-08	
4-LA-104-04	6.02	3.18	—	—	6.06	—	—	—	2006-03-08	
4-LA-104-05	6.02	3.18	—	—	6.04	—	—	—	2006-03-08	
4-LA-104-06	6.02	3.18	—	—	6.10	—	—	—	2006-03-08	
4-LA-104-07	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-104-08	6.02	3.18	—	—	5.94	—	—	—	2006-03-08	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	—	—	8.51	—	—	—	2006-03-08	
4-RA-104-02	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	
4-RA-104-03	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-104-04	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	
4-RA-104-05	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-104-06	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	
4-RA-104-07	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	—	—	3.93	—	—	—	2006-03-08	
2-ST-101-02	3.91	2.29	—	—	3.90	—	—	—	2006-03-08	
2-ST-101-03	3.91	2.29	—	—	3.94	—	—	—	2006-03-08	
2-ST-101-04	3.91	2.29	—	—	3.82	—	—	—	2006-03-08	
2-ST-101-05	3.91	2.29	—	—	3.91	—	—	—	2006-03-08	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	—	—	5.42	—	—	—	2006-03-08	
3-CW-101-02	5.49	2.79	—	—	5.41	—	—	—	2006-03-08	
3-CW-101-03	5.49	2.79	—	—	5.53	—	—	—	2006-03-08	
3-CW-101-04	5.49	2.79	—	—	5.46	—	—	—	2006-03-08	
3-CW-101-05	5.49	2.79	—	—	5.48	—	—	—	2006-03-08	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	—	—	5.50	—	—	—	2006-03-08	
3-CW-102-02	5.49	2.79	—	—	5.51	—	—	—	2006-03-08	
3-CW-102-03	5.49	2.79	—	—	5.53	—	—	—	2006-03-08	
3-CW-102-04	5.49	2.79	—	—	5.51	—	—	—	2006-03-08	

Sign-off	Name	Function	Date	Signature
Prepared	Grace Okafor	UT technician, SNT-TC-1A UT Level II	1996-03-08	_____